

TIØ4145 Corporate Finance

Exercise 3: Capital markets, price of risk, optimal portfolio choice and the CAPM

Deadline: Monday September 15th, 2025, 23:59 – Grading: Passed/Failed

Your answers should be submitted as a PDF file to BlackBoard before the deadline, either written in MS Word, LaTeX or handwritten with a high-quality scan. You are also welcome to upload additional Excel computations if you find it necessary. Guidance to this exercise is given on Thursday September 11th, 8:15 – 10:00 in A32. Solutions to the problems will be posted on BlackBoard after the deadline.

Relevant Curriculum

- Lecture notes from lecture given on September 8th.
- Berk and DeMarzo, Corporate Finance 5th edition: Chapter 11

Tasks

Chapter 11 – Capital markets, price of risk, optimal portfolio choice and the CAPM

1. You own three stocks: 600 shares of Apple Computer, 10,000 shares of Cisco Systems, and 5,000 shares of Colgate-Palmolive. The current share prices and expected returns of Apple, Cisco, and Colgate-Palmolive are, respectively, \$511, \$16, \$97 and 12%, 10%, 8%.
 - a. What are the portfolio weights of the three stocks in your portfolio?
 - b. What is the expected return of your portfolio?
 - c. Suppose the price of Apple stock goes up by \$27, Cisco rises by \$5, and Colgate-Palmolive falls by \$10. What are the new portfolio weights?
 - d. Assuming the stocks' expected returns remain the same, what is the expected return of the portfolio at the new prices?
2. Arbor Systems and Gencore stocks both have a volatility of 43%. Compute the volatility of a portfolio with 50% invested in each stock if the correlation between the stocks is:

- (a) + 1
- (b) 0.50
- (c) 0
- (d) -0.5
- (e) -1.0

In which cases is the volatility lower than that of the original stocks?

3. Consider an equally weighted portfolio of stocks in which each stock has a volatility of 50%, and the correlation between each pair of stocks is 24%.
 - a. What is the volatility of the portfolio as the number of stocks becomes arbitrarily large?
 - b. What is the average correlation of each stock with this large portfolio?

4. You have \$175,000 to invest. You choose to put \$225,000 into the market by borrowing \$50,000.
 - a. If the risk-free interest rate is 6% and the market expected return is 7%, what is the expected return of your investment?
 - b. If the market volatility is 15%, what is the volatility of your investment?

5. In addition to risk-free securities, you are currently invested in the Tanglewood Fund, a broad-based fund of stocks and other securities with an expected return of 12% and a volatility of 25%. Currently, the risk-free rate of interest is 4%. Your broker suggests that you add a venture capital fund to your current portfolio. The venture capital fund has an expected return of 20%, a volatility of 80%, and a correlation of 0.2 with the Tanglewood Fund. Calculate the required return and use it to decide whether you should add the venture capital fund to your portfolio.

6. Your investment portfolio consists of \$13,000 invested in only one stock—Amazon. Suppose the risk-free rate is 6%, Amazon stock has an expected return of 11% and a volatility of 39%, and the market portfolio has an expected return of 10% and a volatility of 16%. Under the CAPM assumptions,
 - a. What alternative investment has the lowest possible volatility while having the same expected return as Amazon? What is the volatility of this investment?

- b. What investment has the highest possible expected return while having the same volatility as Amazon? What is the expected return of this investment?

Exam Question – 2019

You are thinking about investing your money in the stock market, which consists only of three stocks: A, B and C. You know that the economy is expected to behave according to the following table. You believe the likelihood of each scenario is identical (all states of nature have equal probabilities).

State of the economy	R_A	R_B	R_C
Recession	0%	0%	-5%
Normal	10%	20%	10%
Boom	20%	10%	7%

- Calculate the expected returns and the total risk for each of the stocks.
- Calculate the covariances between stocks A and B, stocks A and C, and stocks B and C.
- If you have to form an equally weighted portfolio consisting of two stocks, which two stocks would you put in your portfolio if you only care about risk reduction? Why?
- Suppose now that you are considering creating a portfolio that consists of 60% investment in stock A and 40% investment B. Does this portfolio improve your risk return trade-off relative to the individual investments in stocks A, B and C?
- Does it make sense to add stock C to this portfolio? Why?
- Consider now the market consisting of only two risky stocks A and B with expected returns given above, and a risk-free asset with return equal to 3%. Find the weights of a tangent portfolio in this market and the Sharpe ratio corresponding to this portfolio.

Exam Question – August 2015 [Voluntary, but recommended:)]

You are given the following information about three stocks:

Stock	Expected Return	Standard Deviation
A	0.05	0
B	0.1	0.1
C	0.2	0.35

The correlation between B and C is 0.2.

- a. Suppose you want to invest in any one of the stocks listed above (singly). Can any be recommended?
- b. Now suppose you diversify into two securities. Given all choices, can any portfolio be eliminated? Assume equal weights.

For the next questions (c and d) consider only the two stocks B and C.

- c. Plot, in a mean standard deviation diagram, expected return and standard deviation for portfolios with weights in B of 0, 0.25, 0.5, 0.75 and 1.0. Do this for correlations between the returns of B and C of -0.2, 0 and +0.2.
- d. What does this tell you about how diversification possibilities vary with covariances?

References

Berk, J. and DeMarzo, P. (2017). Corporate Finance, Fifth Edition, Pearson Education, Essex, England.